

Virtual Internship Task Report

Introduction

This document outlines the tasks and responsibilities undertaken as a consultant for a TATA online retail store. The goal is to provide valuable data insights to the CEO and CMO to help them strategically plan for future growth and expansion. Each section details the background information provided for a specific task, followed by a clear description of the task performed and the outcomes achieved.

Task 1: Draft Important Questions

Background Information

- **Key Context:** The business is performing well, and a meeting with the CEO and CMO has been scheduled. The objective is to draft relevant analytics and insights to evaluate current performance and guide future expansion decisions based on available demographic data.
- **Project Goal:** To analyze the major contributing factors to revenue and provide insights for strategic planning and business expansion. The leadership is interested in both operations and marketing perspectives and wants to identify areas of strength and growth opportunities.

Task Performed

- **Specific Actions:** I drafted a set of four questions, differentiating them for the CEO and CMO based on their respective business lenses and anticipating what they need to make effective decisions.
 - **For the CEO (Operations Focus):**
 1. Which products contribute the most to revenue, and how has their performance changed over time? *(identifies best-selling items for planning inventory accordingly)*
 2. What are the key trends in sales performance across different countries, and which regions should we focus on for expansion? *(provides insights into market potential and areas of opportunity)*
 3. Are there any seasonal or monthly patterns in customer purchases that we should consider for demand forecasting? *(for resource allocation and promotional planning)*
 4. How has customer retention evolved over time, and what factors contribute to repeat purchases? *(assesses long-term customer value for loyalty strategies)*
 - **For the CMO (Marketing Focus):**
 1. Which customer segments generate the highest revenue, and what are their

purchasing behaviors? *(Enables targeted marketing campaigns for high-value customers)*

2. What are the most frequently purchased products, and how can we optimize cross-selling and upselling strategies? *(Helps in increasing average order value and maximizing revenue per transaction)*
3. How does customer purchase behavior vary across different countries, and what marketing strategies work best in each region? *(Helps in localizing marketing efforts for better engagement)*
4. What are the key indicators of customer churn, and how can we improve engagement and retention rates? *(Provides actionable insights for reducing churn and improving lifetime value)*

Outcome/Deliverables

- **Deliverable:** A list of eight strategic questions for the CEO and CMO.
- **Results:** This deliverable provides a framework for the analytics to be performed, ensuring the data insights are directly relevant to the executives' strategic goals.

Task 2: Provide Visuals on Metrics

Background Information

- **Key Context:** Senior management wants to understand how their business is performing and what areas are key strengths. They are also focused on identifying opportunities that would lead to growth and generate more revenue in the future. The task requires determining the most suitable visual for four of the eight questions drafted in Task 1.
- **Project Goal:** To provide visuals on key metrics to help the CEO and CMO understand business performance, identify strengths, and pinpoint growth opportunities.

Task Performed

- **Specific Actions:** I selected the best visual representation for each of the four core questions.
 - **Question 1 (CEO):** Time series chart showing monthly revenue for 2011 to identify seasonal trends.
 - **Question 2 (CMO):** A bar chart illustrating the top 10 countries by revenue, with a secondary axis or data labels showing the quantity sold. The UK will be excluded.
 - **Question 3 (CMO):** A descending bar chart of the top 10 customers by revenue to identify the most valuable customers.
 - **Question 4 (CEO):** A map that visually represents product demand across different countries, with high-demand regions highlighted. This provides a single view for identifying expansion opportunities without scrolling.

Outcome/Deliverables

- **Deliverable:** A document outlining the planned visuals for each key business question.
- **Results:** This plan ensures that the subsequent data analysis and visual creation will directly address the executives' needs, making the final presentation highly effective and insightful.

Task 3: Create Visuals and Analysis

Background Information

- **Key Context:** The executives want to understand how to use the data to make more meaningful decisions. They are interested in analyzing trends and breakdowns by different categories to gain clarity on revenue generation and its key drivers.
- **Project Goal:** To create analysis and visuals that answer the CEO and CMO's questions, provide actionable insights, and inform the company's expansion strategy.

Task Performed

- **Specific Actions:** I will create a series of visuals on separate tabs, with each tab named after the corresponding question number.
 - **Question 1:** A line chart displaying revenue over each month in 2011.
 - **Question 2:** A bar chart showing the top 10 countries by revenue (excluding the UK) with the quantity sold.
 - **Question 3:** A bar chart with the top 10 customers ranked by revenue.
 - **Question 4:** A map visual showing product demand across all countries (excluding the UK).
- **Tools/Technologies Used:** Microsoft Power BI and the provided retail dataset.

Outcome/Deliverables

- **Deliverable:** A dashboard or presentation with four separate visuals, each addressing a specific question from the CEO and CMO. Recommendations for increasing sales and expansion opportunities.
- **Results:** The final set of visuals will provide clear, data-driven insights that can be used to inform the executives' decisions regarding business performance, marketing strategies, and future expansion plans.

Task 4: Present Findings to the CEO and CMO

Background Information

- **Key Context:** The executives are keen to understand the thought process behind the analysis and the key conclusions drawn from the data. The presentation needs to be compelling and directly address the four questions from previous tasks, with a special emphasis on guiding expansion strategy.
- **Project Goal:** To present a clear, straightforward explanation of the data cleanup, analysis, and visualization process to the CEO and CMO, focusing on lucrative expansion opportunities.

Task Performed

- **Specific Actions:** I developed a script for a presentation to be delivered to the CEO and CMO. The script explains the data-handling process and presents the analysis for each of the four questions.
- **Script:**

Good Afternoon,

I'm Michael Mensah, and I'm excited to share some insights about your business. Thank you for providing the guiding questions. It was helpful to see what types of insights you are looking to gain from the data. I hope you find the analysis compelling and helpful as you make decisions regarding future business opportunities.

First off, I want to assure you that I've provided the most up to date and error free analysis. After I loaded the data into my software, I scrubbed any records that have negative quantities and unit price, as these records needed to be removed in order to provide helpful analysis.

As for your first question, the CEO has requested a trend of the revenue to see if there is any seasonality in the store sales. My analysis shows that there are some months of the year where exceptional growth is witnessed. The data shows that the revenue in the first 8 months is fairly constant as the average revenue generated for these 8 months is around \$685k. The increase in revenue starts in the month of September, where the revenue increases by 40% over the previous month. This trend continues till the month of November where it reached 1.5 million USD, the highest during the entire year. The data is incomplete for the month of December, therefore, no conclusion can be drawn from it, unfortunately. This analysis shows that the retail store sales are impacted by the seasonality which usually occurs in the last 4 months of the year.

The second visual shows how the top 10 countries which have opportunities for growth are performing. This data does not include the UK as the country already has high demand and I've been told you're more focused on the countries where demand can be increased. The analysis shows that countries such as the

Netherlands, Ireland, Germany and **France** have high volumes of units bought and revenue generated. I would suggest that these countries should be focused on to ensure that measures are taken to capture these markets even more.

The third analysis has been performed on the top 10 customers who have purchased the most from the store. The data shows that there is not much of a difference between the purchases made by the top 10 customers. The highest revenue generating customer only purchased 17% more than the 2nd highest which shows that the business is not relying only on a few customers to generate the revenue. This shows that the bargaining power of customers is low and the business is in a good position.

Finally, the map chart shows the regions that have generated the most revenue compared with the regions that have not. It can be seen that apart from the UK, countries such as the **Netherlands, Ireland, Germany, France**, and **Australia** are generating high revenue and the company should invest more in these areas to increase demand for products. The map also shows that most of the sales are only in the European region with very few in the American region. **Africa** and **Asia** do not have any demand for the products, along with **Russia**. A new strategy targeting these areas has the potential to boost sales revenues and profitability.

Thanks so much for your time. If you have any questions about the analysis or would like to see anything additional after you've had time to digest this information, I'd be happy to develop that for you.

Outcome/Deliverables

- **Deliverable:** A comprehensive presentation that explains the methodology, analysis, and key findings.
- **Results:** The presentation provides a clear, structured narrative for presenting the insights to the executives, directly answering their questions and guiding them toward potential expansion opportunities.